# Toward an Antimatter Periodic Table: A Preparedness Framework

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Version v1.0.0 – 26 April 2026

**Version:** v1.0.0
**Release date:** 26 April 2026
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**Keywords:** antimatter; periodic table; CPT symmetry; antihydrogen; anti-atoms; evidence status; preparedness framework

**Abstract**

Antimatter does not presently require a new chemical periodic law. Under the standard assumptions of the CPT theorem, isolated anti-atoms are expected to share the internal spectral and orbital structure of their matter counterparts. The empirical evidence, however, is uneven. Antihydrogen is the only pure anti-atomic system that has been produced, trapped, spectroscopically characterized, laser cooled, and gravitationally tested. Antihelium-nucleus evidence is established for antihelium nuclei, not for a neutral antihelium atom.

This work therefore proposes a CPT-isomorphic evidence-status and preparedness framework rather than an alternative periodic law. The framework separates theoretically expected periodicity from the experimental realization of individual antimatter systems. Its main representation is a split-cell table that combines classical periodic structure with an explicit evidence layer. This is a representation and evidence-status framework, not a claim of a new periodic law or established antimatter chemistry. Residual notation is reserved for measured pulls or explicitly defined hypothetical model values in named observables. The intended contribution is a cautious representation layer for future anti-atom, anti-ion, anti-molecule, and astrophysical antimatter searches.

## Contents

**1 Aim and Boundary** **3**

**2 Nomenclature** **3**
2.1 Names and Working Notation . . . 3

**3 Current State of Evidence** **4**
3.1 Antihydrogen as the Experimental Anchor . . . 4
3.2 Antinuclei: Antihelium Nuclei and Hypernuclei . . . 5
3.3 Anti-Ions, Anti-Molecules, and Chemistry . . . 5
3.4 Precision Constraints . . . 5
3.5 Astrophysical Antimatter . . . 5

**4 Physical Structure Model** **6**
4.1 CPT-Isomorphic Mapping . . . 6
4.2 Reduced Mass and Spectral Shifts . . . 6
4.3 When the Periodic Table Stops Being the Right Representation . . . 7

**5 Status Vector** **7**

**6 Residual Notation** **8**

**7 Source Basis for the Tables** **8**

**8 Representation Hierarchy** **9**
8.1 Split-Cell Antimatter Periodic Table . . . . . 9
8.2 Comparison . . . . . 11

**9 Open Scientific Domains** **11**
9.1 Experimental Realization . . . . . 11
9.2 Anti-Molecules and Reaction Chemistry . . . . . 12
9.3 Astrophysical Antimatter . . . . . 12
9.4 Superheavy Elements . . . . . 12

**10 Hypothetical Deviation Layer** **12**

**11 Conclusion** **12**

# 1 Aim and Boundary

The phrase "antimatter periodic table" is scientifically useful only if it is not confused with a speculative new chemistry. The central thesis is:

> An antimatter periodic table is not a new periodic law under present standard physics. It is a CPT-isomorphic evidence-status and preparedness framework that keeps the classical orbital structure separate from the empirical realization of the corresponding antimatter systems.

The term CPT-isomorphic is used here conditionally: it refers to matched isolated systems under the standard assumptions of CPT-invariant local relativistic quantum field theory. Preparedness means that the framework records which systems are structurally expected, experimentally open, or tied to future search programs.

This distinction matters. Under local, Lorentz-invariant, unitary quantum field theory, CPT symmetry implies equal masses and lifetimes for particles and antiparticles, with opposite charges. A genuine mass or spectral difference would not be a decorative table feature; it would point either to physics beyond the Standard Model or to uncontrolled systematic effects. Greenberg showed that CPT violation in interacting local theories implies Lorentz violation, so particle-antiparticle mass splittings would not be harmless in such a setting [3].

The statement is deliberately conditional. CPT invariance rests on the usual assumptions of local relativistic quantum field theory and is reviewed, for example, by Lehnert [4]; the older TCP/CPT theorem literature is part of that foundation [2]. The framework below does not propose a new test theory for CPT violation. It uses CPT-based expectations as the structural reference and keeps experimental evidence, open systems, and hypothetical residuals in separate layers.

The paper addresses four questions:

1. Which antimatter systems are directly realized at present?
2. How can the distinction between CPT-expected and experimentally confirmed be encoded?
3. Which periodic-table representation is appropriate for which state of knowledge?
4. How should the representation change if a genuine matter-antimatter asymmetry is measured?

# 2 Nomenclature

## 2.1 Names and Working Notation

IUPAC defines atomic number as proton number $Z$ and ratifies names and symbols for ordinary chemical elements [5, 7]. There is no separate IUPAC series of official antimatter element names. In physics, a practical notation is used: the prefix "anti-" in the name and an overbar in the symbol, for example antihydrogen ($\overline{\mathrm{H}}$, plain-text: anti-H). A neutral antihelium atom would be written $\overline{\mathrm{He}}$ (plain-text: anti-He), but such a neutral anti-atom is not established in the cited literature. This follows the particle-antiparticle language rooted in relativistic quantum theory [1].

This framework uses

$$E_Z \longmapsto \overline{\mathrm{E}}_Z, \qquad N_{\bar{p}} = Z. \tag{1}$$

The antinuclear charge is $-Ze$, but the table position remains indexed by the positive integer $Z$. The antimatter side is therefore not represented by negative atomic numbers and not by $Z \mapsto -Z$.

Table 1: Terminology for an antimatter periodic-table framework.

| Object | Preferred term | Notation | Status / comment |
|---|---|---|---|
| Antiproton | antiproton | $\bar{p}$ | Established antiparticle; not a chemical element. |
| Positron | positron / antielectron | $e^+$ | IUPAC defines the positron as the positively charged electron [6]. |
| Neutral antihydrogen atom | antihydrogen | $\overline{\mathrm{H}}$ (anti-H) | Pure anti-atom made from $\bar{p}$ and $e^+$; experimental anchor of the framework [8, 9, 10, 11, 14]. |
| Antihydrogen ion | positive antihydrogen ion | $\overline{\mathrm{H}}^+$ (anti-H$^+$) | Antiproton plus two positrons; target system in GBAR-style gravity programs and experimentally harder than neutral $\overline{\mathrm{H}}$; the cited GBAR work reports neutral antihydrogen production through positronium, not an established positive antihydrogen-ion observation [25]. |
| Antihelium nucleus | antihelium-3 or antihelium-4 nucleus | $\overline{{}^3\mathrm{He}}$, $\overline{{}^4\mathrm{He}}$ | Antinucleus observed; not the same as a neutral antihelium atom [28]. |
| Antiprotonic helium | antiprotonic helium | $\bar{p}\mathrm{He}^+$ | Exotic mixed system containing an ordinary helium nucleus, an electron, and an antiproton; not an antihelium nucleus and not a neutral antihelium atom [30]. |
| Generic antimatter counterpart notation | anti- + element name | $\overline{\mathrm{E}}$ (anti-E) or $\overline{\mathrm{X}}$ (anti-X) | Informal working notation, not a new IUPAC naming series. For multi-letter symbols, the overbar covers the whole symbol, as in $\overline{\mathrm{He}}$ (anti-He). |

# 3 Current State of Evidence

## 3.1 Antihydrogen as the Experimental Anchor

Antihydrogen is the only pure anti-atom that has been systematically studied. Cold antihydrogen was first produced and detected by ATHENA and ATRAP in 2002 [8, 9]. ALPHA reported trapped antihydrogen in 2010 and confinement for 1,000 seconds in 2011 [10, 11]. Subsequent ALPHA measurements observed and characterized the $1S$–$2S$ transition, the $1S$–$2P$ Lyman-alpha transition, hyperfine spectrum, fine structure, Lamb-shift-related intervals, laser cooling, and hyperfine-resolved $1S$–$2S$ spectroscopy of trapped antihydrogen [12, 13, 14, 15, 16, 17, 18]. These measurements provide direct spectroscopy tests of the CPT-expected isomorphism. The 2023 ALPHA-g measurement found that released antihydrogen atoms fall in a way compatible with gravitational attraction toward Earth; it is a finite-precision result for neutral antihydrogen in the terrestrial field, not a general high-precision equivalence-principle test for all antimatter. Repulsive “antigravity” for neutral antihydrogen under those conditions is excluded at the present

experimental sensitivity [19]. In 2025, ALPHA also reported an eightfold increase in trapping rate through $Be^+$-assisted positron cooling and the accumulation of more than 15,000 antihydrogen atoms in less than seven hours [20].

Antihydrogen therefore has a special status:

$$\mathbf{S}(\overline{\mathrm{H}}) = (3, 3, 3, 1, 0), \tag{2}$$

where the vector components are defined in Section 5. The chemistry component remains theoretical because reaction chemistry and stable anti-molecule data are not experimentally catalogued at the same level as spectroscopy.

### 3.2 Antinuclei: Antihelium Nuclei and Hypernuclei

The STAR Collaboration observed the antihelium-4 nucleus, the antimatter counterpart of the alpha particle [28]. In 2024, STAR reported a more complex antimatter hypernucleus [29]. These results show that heavy antimatter nuclear systems can be produced in high-energy collisions.

They do not imply that a neutral antihelium atom has been produced or chemically studied. For a periodic-table framework, the distinction is essential:

$$\overline{{}^4\mathrm{He}}_{\mathrm{nucleus}} \neq \overline{\mathrm{He}}_{\mathrm{atom}} = \left(\overline{{}^4\mathrm{He}}_{\mathrm{nucleus}}, e^+, e^+\right). \tag{3}$$

### 3.3 Anti-Ions, Anti-Molecules, and Chemistry

The chemistry layer is presently a theoretical and prospective preparedness layer, not an empirical catalogue of established antimatter chemistry. Zammit et al. discuss formation pathways for antihydrogen and for anionic, cationic, and molecular antimatter species through collision-driven and radiation-driven processes [27]. Related work has proposed laser-driven production of the antihydrogen molecular ion [26]. These references support the direction of the framework: a table should not imply that anti-molecules have already been chemically catalogued. It should show which species are theoretically prepared and experimentally open.

### 3.4 Precision Constraints

The question "what if the mass differs?" is legitimate but strongly constrained. BASE compared the antiproton-to-proton charge-to-mass ratio with a fractional uncertainty of 16 parts per trillion [21]. Antiprotonic helium provides additional precision for the antiproton-to-electron mass ratio [30]. These measurements do not replace every possible anti-atom test, but they tightly constrain simple mass or charge deviations.

### 3.5 Astrophysical Antimatter

The cosmic matter-antimatter asymmetry remains open: the observable universe is overwhelmingly matter dominated. Baryogenesis reviews describe why the observed asymmetry requires physics beyond a naive matter-antimatter balance [33, 34]. Large nearby antimatter domains are strongly constrained by gamma-ray and cosmic-microwave-background considerations, and Fermi-LAT source-catalog analyses place limits on nearby antistar populations [35, 36]. Cosmic antihelium-nucleus or antideuteron signatures would therefore be astrophysically important. GAPS describes a low-energy antinuclei search space where standard astrophysical backgrounds are expected to be small and dark-matter scenarios provide additional motivation [31]; ALICE absorption measurements also improve the propagation inputs for antihelium-nucleus searches in space [32]. Such programs belong in an astrophysical evidence layer, not in chemical periodicity, and they concern antinuclei rather than neutral antihelium atoms.

# 4 Physical Structure Model

## 4.1 CPT-Isomorphic Mapping

A neutral atom with nucleus $N_Z$ and $Z$ electrons maps to a neutral anti-atom with antinucleus $\overline{N}_Z$ and $Z$ positrons:

$$\mathcal{A}_{Z,A} = \left(N_{Z,A}, e_1^-, \ldots, e_Z^-\right) \xrightarrow{CPT} \overline{\mathcal{A}}_{Z,A} = \left(\overline{N}_{Z,A}, e_1^+, \ldots, e_Z^+\right). \tag{4}$$

An idealized nonrelativistic electronic Hamiltonian for matter is

$$H_Z = \sum_{i=1}^{Z} \left[\frac{\mathbf{p}_i^2}{2m_e} - \frac{Ze^2}{4\pi\varepsilon_0 r_i}\right] + \sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 r_{ij}} + H_{\text{rel}} + H_{\text{QED}} + H_{\text{nuc}}. \tag{5}$$

For the anti-atom, electrons are replaced by positrons and the nucleus by the antinucleus. Under charge conjugation, the relevant charge products remain invariant:

$$(+Ze)(-e) = (-Ze)(+e) = -Ze^2, \qquad (-e)(-e) = (+e)(+e) = +e^2. \tag{6}$$

The charge-product argument justifies the nonrelativistic Coulomb analogy. Full equality of matter and antimatter spectra is a stronger statement, tied to the CPT theorem and to matching internal quantum numbers, boundary conditions, and external fields. For many-electron and high-$Z$ systems this is a structural heuristic, not a derived many-body theorem for every chemical property; correlation, relativistic, QED, recoil, and nuclear-size effects must be carried through in any detailed calculation. For many-electron systems, the framework encodes expected structural correspondence, not computed chemical observables. In that idealized setting,

$$\text{Spec}\,(H_Z) = \text{Spec}\left(\overline{H}_Z\right) \quad \text{for matched isolated systems.} \tag{7}$$

Chemical periodicity is therefore structurally preserved as long as CPT holds, the relevant internal states are matched, and no asymmetric environmental interaction changes the anti-atom.

## 4.2 Reduced Mass and Spectral Shifts

For hydrogen-like systems, the reduced mass is

$$\mu = \frac{m_\ell M_N}{m_\ell + M_N}. \tag{8}$$

The leading binding energy is

$$E_n \simeq -\frac{\mu c^2}{2n^2}(Z\alpha)^2 + \Delta_{\text{fs}} + \Delta_{\text{Lamb}} + \Delta_{\text{hfs}} + \Delta_{\text{recoil}}. \tag{9}$$

If hypothetical deviations are parameterized as

$$m_{e^+} = m_e(1+\delta_e), \qquad M_{\text{antiN}} = M_N(1+\delta_{\text{antiN}}), \tag{10}$$

then the leading reduced-mass contribution is

$$\frac{\delta\mu}{\mu} \simeq \frac{M_N}{m_e + M_N}\delta_e + \frac{m_e}{m_e + M_N}\delta_{\text{antiN}}. \tag{11}$$

A positron-electron mass difference would enter anti-atomic spectra almost directly. An antinuclear mass difference is suppressed in electronic energy levels by $m_e/(m_e + M_N)$, but it can still appear in precision and recoil terms. For chemical bonds, characteristic scales shift as

$$a_0 \propto \frac{1}{\mu\alpha}, \qquad E_{\text{ion}} \propto \mu\alpha^2. \tag{12}$$

A small mass difference would first shift bond lengths, ionization energies, and isotope shifts. It would not automatically rewrite the group structure. These reduced-mass equations are intended for hydrogen-like scale estimates. They are not a proof of unchanged many-electron periodic ordering across the whole table.

### 4.3 When the Periodic Table Stops Being the Right Representation

The classical table remains appropriate while the ordering of relevant levels is preserved. Here "mat" denotes the matter system and "anti" denotes the corresponding antimatter system in plain-text-safe indices. For each pair of compared levels define

$$\Delta\epsilon_{\text{mat}} = \epsilon^{\text{mat}}_{n\ell j}(Z) - \epsilon^{\text{mat}}_{n'\ell' j'}(Z), \qquad \Delta\epsilon_{\text{anti}} = \epsilon^{\text{anti}}_{n\ell j}(Z) - \epsilon^{\text{anti}}_{n'\ell' j'}(Z). \tag{13}$$

The ordering condition is

$$\text{sgn}(\Delta\epsilon_{\text{mat}}) = \text{sgn}(\Delta\epsilon_{\text{anti}}). \tag{14}$$

It becomes a poor primary representation only if a deviation changes orbital ordering or bonding topology:

$$\exists\,(n\ell j, n'\ell' j') : \quad \text{sgn}(\Delta\epsilon_{\text{mat}}) \neq \text{sgn}(\Delta\epsilon_{\text{anti}}). \tag{15}$$

In that case, the appropriate representation would be a residual or phase-diagram map in variables such as

$$(Z, A, q, n, \ell, j; \delta_m, \delta_q, \delta_g, \Delta E, \Delta_{\text{chem}}). \tag{16}$$

## 5 Status Vector

A single confidence value is too coarse for antimatter. It collapses nuclear evidence, neutral anti-atoms, spectroscopy, chemistry, and astrophysical searches into one number. The framework therefore uses an evidence vector:

$$\mathbf{S}(\overline{\text{E}}_Z) = (S_{\text{nuc}}, S_{\text{atom}}, S_{\text{spec}}, S_{\text{chem}}, S_{\text{astro}})\,. \tag{17}$$

Each component uses four levels:

| Level | Name | Meaning |
|---|---|---|
| 0 | open | No direct evidence in the cited literature. |
| 1 | theoretically expected | Inferred from CPT, QED, nuclear physics, or known matter chemistry. |
| 2 | observed | Antinucleus, anti-atom, or species produced or indirectly established. |
| 3 | precisely characterized | Spectroscopy, gravity, mass comparison, reaction rate, or other quantitative data exist. |

Nomenclature is handled separately in Table 1; it is not scored on the same evidence scale as spectroscopy, chemistry, or nuclear production.

| Object | $S_{\text{nuc}}$ | $S_{\text{atom}}$ | $S_{\text{spec}}$ | $S_{\text{chem}}$ | $S_{\text{astro}}$ |
|---|---|---|---|---|---|
| $\overline{\text{H}}$ | 3 | 3 | 3 | 1 | 0 |
| $\overline{\text{He}}$ | 2 | 0 | 0 | 1 | 1 |
| $\overline{\text{Li}}$ to $\overline{\text{Og}}$ | 1 | 0 | 0 | 1 | 0 |
| cosmic antinuclei | 1 | 0 | 0 | 0 | 1 |

The level-1 entries for higher neutral anti-atoms and prospective antimatter chemistry mean that the structures are theoretically expected under the framework. They do not mean that neutral anti-atoms, anti-molecules, or reaction networks have been experimentally realized.

## 6 Residual Notation

If a genuine matter-antimatter difference is searched for or measured, it should not be written as a new table position or as a negative atomic number. For each observable $O$, use a dimensionless pull:

$$\rho_O(\overline{\mathrm{E}}_Z) = \frac{O_{\mathrm{mat}}(Z) - O_{\mathrm{anti}}(Z)}{\sigma_O(Z)}. \tag{18}$$

Here $\sigma_O(Z)$ is the standard uncertainty of the matter-antimatter comparison. In the absence of a published covariance matrix, it should be the quadrature combination of the antimatter measurement uncertainty, the matter or reference uncertainty, and relevant systematic terms; if correlations are available, the published covariance treatment should be used instead. Residual notation is reserved for measured pulls or explicitly defined hypothetical model values. The statement $\rho_O$ compatible with zero denotes CPT-compatible equality within finite uncertainty, not an exact zero residual. A significant negative residual $\rho_O < 0$ or positive residual $\rho_O > 0$ belongs in a residual representation. Missing data are not shifted; they remain marked as unknown. This prevents absence of evidence from being drawn as a physical deviation.

The compact notation is

$$\overline{\mathrm{E}}_Z[O : \rho_O], \tag{19}$$

for example $\overline{\mathrm{H}}[1S\text{–}2S : \rho_{1S\text{–}2S}]$ (anti-H, 1S–2S) when a measured pull is being discussed. For the present framework table, the measured antihydrogen 1S–2S comparison is described in words as compatible with zero residual within uncertainty. For the $\overline{\mathrm{He}}$ (anti-He) entry, where only antihelium nucleus evidence exists, the marker remains $N$ for antinucleus-only status. For higher unrealized anti-atoms, it remains ?.

## 7 Source Basis for the Tables

The classical table geometry, element symbols, and positive $Z$ indexing follow IUPAC definitions of atomic number and currently named elements [5, 7]. The antimatter status markers are evidence classifications derived from the cited literature and from the repository data table `data/element_status_table.csv`. A marker such as ? means that no neutral anti-atom realization is established in the cited literature. It is not a proof that the corresponding anti-system cannot exist.

Table 2: Source provenance for the evidence classes used in the tables.

| **Table item** | **Source basis** | **Interpretation in this framework** |
|---|---|---|
| Classical element positions | IUPAC atomic number and element-name sources [5, 7] | The table index is the positive atomic number $Z$; antimatter is not represented by negative $Z$. |

| Table item | Source basis | Interpretation in this framework |
|---|---|---|
| $\overline{\mathrm{H}}$ (anti-H) marker = | ATHENA, ATRAP, and ALPHA antihydrogen production, trapping, spectroscopy, gravity, laser-cooling, and trapping-rate results [8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] | Antihydrogen is the experimental anti-atom anchor and is compatible with CPT expectations within measured uncertainty. |
| $\overline{\mathrm{He}}$ (anti-He) marker $N$ | STAR antihelium nucleus and antimatter-hypernucleus observations [28, 29] | The $\overline{\mathrm{He}}$ marker denotes antihelium-nucleus evidence only; neutral antihelium atoms are not established by these sources. |
| Residual constraints | BASE antiproton charge-to-mass comparison, antiprotonic-helium spectroscopy, antihydrogen precision tests, and SME-motivated hydrogen-antihydrogen comparisons [21, 30, 14, 22, 23] | Residual entries are tied to named measured observables or clearly hypothetical coefficients, not to missing evidence. |
| Higher neutral anti-atoms and anti-molecules | CPT-based structure model plus available antihydrogen-chemistry preparedness literature [26, 27] | These entries are theoretically expected or experimentally open; no neutral anti-atom realization beyond $\overline{\mathrm{H}}$ is established in the cited literature. |
| $\overline{\mathrm{H}}$ (anti-H) beam and positive ion pathway | ASACUSA antihydrogen source work and GBAR neutral-antihydrogen production through positronium [24, 25] | These works support future spectroscopy and gravity programs; they do not establish a positive antihydrogen ion, heavier neutral anti-atoms, or antimatter chemistry. |

# 8 Representation Hierarchy

A useful representation must respect known periodicity and avoid overstating evidence. The split-cell table is the main figure for the current state of knowledge. The classical table is the implicit reference. Mirror layouts and residual maps are useful only as supplementary or didactic views: a mirror layout visualizes charge conjugation, not negative atomic numbers, and a residual map compares named observables rather than chemical groups.

## 8.1 Split-Cell Antimatter Periodic Table

## Split-cell evidence-status table

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 H / $\overline{\text{H}}$ = | | | | | | | | | | | | | | | | | 2 He / $\overline{\text{He}}$ N |
| 3 Li / $\overline{\text{Li}}$ ? | 4 Be / $\overline{\text{Be}}$ ? | | | | | | | | | | | 5 B / $\overline{\text{B}}$ ? | 6 C / $\overline{\text{C}}$ ? | 7 N / $\overline{\text{N}}$ ? | 8 O / $\overline{\text{O}}$ ? | 9 F / $\overline{\text{F}}$ ? | 10 Ne / $\overline{\text{Ne}}$ ? |
| 11 Na / $\overline{\text{Na}}$ ? | 12 Mg / $\overline{\text{Mg}}$ ? | | | | | | | | | | | 13 Al / $\overline{\text{Al}}$ ? | 14 Si / $\overline{\text{Si}}$ ? | 15 P / $\overline{\text{P}}$ ? | 16 S / $\overline{\text{S}}$ ? | 17 Cl / $\overline{\text{Cl}}$ ? | 18 Ar / $\overline{\text{Ar}}$ ? |
| 19 K / $\overline{\text{K}}$ ? | 20 Ca / $\overline{\text{Ca}}$ ? | 21 Sc / $\overline{\text{Sc}}$ ? | 22 Ti / $\overline{\text{Ti}}$ ? | 23 V / $\overline{\text{V}}$ ? | 24 Cr / $\overline{\text{Cr}}$ ? | 25 Mn / $\overline{\text{Mn}}$ ? | 26 Fe / $\overline{\text{Fe}}$ ? | 27 Co / $\overline{\text{Co}}$ ? | 28 Ni / $\overline{\text{Ni}}$ ? | 29 Cu / $\overline{\text{Cu}}$ ? | 30 Zn / $\overline{\text{Zn}}$ ? | 31 Ga / $\overline{\text{Ga}}$ ? | 32 Ge / $\overline{\text{Ge}}$ ? | 33 As / $\overline{\text{As}}$ ? | 34 Se / $\overline{\text{Se}}$ ? | 35 Br / $\overline{\text{Br}}$ ? | 36 Kr / $\overline{\text{Kr}}$ ? |
| 37 Rb / $\overline{\text{Rb}}$ ? | 38 Sr / $\overline{\text{Sr}}$ ? | 39 Y / $\overline{\text{Y}}$ ? | 40 Zr / $\overline{\text{Zr}}$ ? | 41 Nb / $\overline{\text{Nb}}$ ? | 42 Mo / $\overline{\text{Mo}}$ ? | 43 Tc / $\overline{\text{Tc}}$ ? | 44 Ru / $\overline{\text{Ru}}$ ? | 45 Rh / $\overline{\text{Rh}}$ ? | 46 Pd / $\overline{\text{Pd}}$ ? | 47 Ag / $\overline{\text{Ag}}$ ? | 48 Cd / $\overline{\text{Cd}}$ ? | 49 In / $\overline{\text{In}}$ ? | 50 Sn / $\overline{\text{Sn}}$ ? | 51 Sb / $\overline{\text{Sb}}$ ? | 52 Te / $\overline{\text{Te}}$ ? | 53 I / $\overline{\text{I}}$ ? | 54 Xe / $\overline{\text{Xe}}$ ? |
| 55 Cs / $\overline{\text{Cs}}$ ? | 56 Ba / $\overline{\text{Ba}}$ ? | 71 Lu / $\overline{\text{Lu}}$ ? | 72 Hf / $\overline{\text{Hf}}$ ? | 73 Ta / $\overline{\text{Ta}}$ ? | 74 W / $\overline{\text{W}}$ ? | 75 Re / $\overline{\text{Re}}$ ? | 76 Os / $\overline{\text{Os}}$ ? | 77 Ir / $\overline{\text{Ir}}$ ? | 78 Pt / $\overline{\text{Pt}}$ ? | 79 Au / $\overline{\text{Au}}$ ? | 80 Hg / $\overline{\text{Hg}}$ ? | 81 Tl / $\overline{\text{Tl}}$ ? | 82 Pb / $\overline{\text{Pb}}$ ? | 83 Bi / $\overline{\text{Bi}}$ ? | 84 Po / $\overline{\text{Po}}$ ? | 85 At / $\overline{\text{At}}$ ? | 86 Rn / $\overline{\text{Rn}}$ ? |
| 87 Fr / $\overline{\text{Fr}}$ ? | 88 Ra / $\overline{\text{Ra}}$ ? | 103 Lr / $\overline{\text{Lr}}$ ? | 104 Rf / $\overline{\text{Rf}}$ ? | 105 Db / $\overline{\text{Db}}$ ? | 106 Sg / $\overline{\text{Sg}}$ ? | 107 Bh / $\overline{\text{Bh}}$ ? | 108 Hs / $\overline{\text{Hs}}$ ? | 109 Mt / $\overline{\text{Mt}}$ ? | 110 Ds / $\overline{\text{Ds}}$ ? | 111 Rg / $\overline{\text{Rg}}$ ? | 112 Cn / $\overline{\text{Cn}}$ ? | 113 Nh / $\overline{\text{Nh}}$ ? | 114 Fl / $\overline{\text{Fl}}$ ? | 115 Mc / $\overline{\text{Mc}}$ ? | 116 Lv / $\overline{\text{Lv}}$ ? | 117 Ts / $\overline{\text{Ts}}$ ? | 118 Og / $\overline{\text{Og}}$ ? |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 57 La / $\overline{\text{La}}$ ? | 58 Ce / $\overline{\text{Ce}}$ ? | 59 Pr / $\overline{\text{Pr}}$ ? | 60 Nd / $\overline{\text{Nd}}$ ? | 61 Pm / $\overline{\text{Pm}}$ ? | 62 Sm / $\overline{\text{Sm}}$ ? | 63 Eu / $\overline{\text{Eu}}$ ? | 64 Gd / $\overline{\text{Gd}}$ ? | 65 Tb / $\overline{\text{Tb}}$ ? | 66 Dy / $\overline{\text{Dy}}$ ? | 67 Ho / $\overline{\text{Ho}}$ ? | 68 Er / $\overline{\text{Er}}$ ? | 69 Tm / $\overline{\text{Tm}}$ ? | 70 Yb / $\overline{\text{Yb}}$ ? |
| 89 Ac / $\overline{\text{Ac}}$ ? | 90 Th / $\overline{\text{Th}}$ ? | 91 Pa / $\overline{\text{Pa}}$ ? | 92 U / $\overline{\text{U}}$ ? | 93 Np / $\overline{\text{Np}}$ ? | 94 Pu / $\overline{\text{Pu}}$ ? | 95 Am / $\overline{\text{Am}}$ ? | 96 Cm / $\overline{\text{Cm}}$ ? | 97 Bk / $\overline{\text{Bk}}$ ? | 98 Cf / $\overline{\text{Cf}}$ ? | 99 Es / $\overline{\text{Es}}$ ? | 100 Fm / $\overline{\text{Fm}}$ ? | 101 Md / $\overline{\text{Md}}$ ? | 102 No / $\overline{\text{No}}$ ? |

**Block:** s p d f Marker symbols are primary; colors are redundant; no negative Z.

**Marker:** = compatible within finite uncertainty, not exact equality.
N antinucleus only; ? experimentally open; R named residual only.

**Anti status:** Atom Nucleus Open

**Status summary:** anti-H: atom + spectroscopy + gravity.
anti-He: antinucleus only, no neutral anti-He atom.
$Z \geq 3$: neutral anti-atom open; marker symbols work in grayscale.

Figure 1: Split-cell antimatter periodic-table framework. Classical positions follow positive atomic number $Z$; antimatter is not represented by negative $Z$. Markers are evidence classifications: = denotes compatibility with CPT expectations within finite uncertainty, not exact equality; $N$ denotes antihelium-nucleus evidence only; ? denotes experimentally open realization; and $R$ is reserved for named residual observables only.

The split-cell representation is the main representation for the current framework. It separates the classical orbital-block structure from the experimental antimatter status. It makes four statements visible at once:

- the periodic law is not reinvented;
- $\overline{\mathrm{H}}$ (anti-H) is experimentally better established than all other anti-atoms;
- $\overline{\mathrm{He}}$ (anti-He) has antihelium-nucleus status but no neutral anti-atom status;
- for $Z \geq 3$, the position is structurally expected but not experimentally realized as a neutral anti-atom.

$$C : (Z, e^-, N_Z) \mapsto (Z, e^+, \overline{N}_Z), \tag{20}$$

not as $Z \mapsto -Z$. Residual notation, when used, should compare measured pulls such as $m/q$, the $1S$–$2S$ frequency, hyperfine structure, gravitational acceleration, ionization energy, or a reaction rate. It should not be used to visualize missing data.

### 8.2 Comparison

Table 3: Comparison of representation choices.

| **View** | **Strength** | **Risk** | **Use** |
|---|---|---|---|
| Classical reference | Clean block structure and known chemistry | No antimatter information | Introduction and calibration |
| Split-cell table | Separates structure and evidence in each cell | Denser than a standard table | Main figure for current knowledge |
| Mirror representation | Intuitive matter-antimatter juxtaposition | Can suggest negative atomic numbers | Teaching and overview only |
| Residual map | Handles genuine deviations cleanly | Not a periodic table in the chemical sense | Supplementary CPT, mass, spectral, and gravity residuals |
| Superheavy extension | Addresses relativistic heavy-element modeling | Model-dependent and not anti-atom evidence | Separate theory layer |

## 9 Open Scientific Domains

### 9.1 Experimental Realization

The largest uncertainty is not the orbital structure. It is the production, cooling, trapping, and measurement of neutral anti-atoms beyond $\overline{\mathrm{H}}$. An anti-atom with $Z \geq 2$ requires an antinucleus and a controlled positron shell. Antinuclei from high-energy collisions are rare and usually not cold, neutral, or long-lived enough for chemistry.

### 9.2 Anti-Molecules and Reaction Chemistry

Standard physics expects isolated antimatter chemistry to be isomorphic to matter chemistry, but present antimatter-chemistry statements are largely prospective and theoretical. Laboratory chemistry is not just a Hamiltonian. It requires sources, traps, cooling, density, reaction partners, and wall-free conditions. The open quantities include

$$\sigma_{\text{form}}, \quad k_{\text{reaction}}(T), \quad \tau_{\text{trap}}, \quad n_{\overline{\text{M}}}, \quad \Gamma_{\text{ann}}. \tag{21}$$

### 9.3 Astrophysical Antimatter

Macroscopic antimatter domains would be cosmological evidence, not new chemistry. Antistars or antimatter clusters would change how baryogenesis or transport models are assessed, and they are constrained by the absence of expected annihilation signatures on accessible scales and by gamma-ray source-catalog searches [35, 36]. In this framework, that evidence belongs in $S_{\text{astro}}$.

### 9.4 Superheavy Elements

The IUPAC table currently ends at $Z = 118$ [7]. For $Z > 118$, chemical periodicity is theoretically modeled but not experimentally established as element chemistry. Pyykko proposed an extension up to $Z \leq 172$ based on relativistic Dirac-Fock calculations [37]. This is relevant because relativistic, QED, and nuclear-volume effects grow at large $Z$, but it is a separate theory layer:

$$\mathbf{S}_{\text{total}} = \mathbf{S}_{\text{anti}} \oplus \mathbf{S}_{\text{superheavy}}. \tag{22}$$

## 10 Hypothetical Deviation Layer

A hypothetical matter-antimatter difference must be tied to named observables. For a hydrogen-like system define fractional mass and charge-magnitude parameters as

$$\delta_{\text{antiN}} = \frac{M_{\text{antiN}} - M_N}{M_N}, \qquad \delta_e = \frac{m_{e^+} - m_e}{m_e}, \qquad \eta_{\text{antiN}} = \frac{|q_{\text{antiN}}| - |q_N|}{|q_N|}, \qquad \eta_{e^+} = \frac{|q_{e^+}| - |q_e|}{|q_e|}. \tag{23}$$

The leading reduced-mass response is already given in Eq. (11). Charge-magnitude terms, nuclear-size terms, QED terms, or Standard-Model Extension coefficients can enter named observables, but they must be defined within the specific model and uncertainty budget being tested [22, 23]. This manuscript therefore treats SME-style terms as hypothetical or externally constrained inputs, not as measured deviations in the table.

As a practical rule, the split-cell table remains the preferred representation while level ordering and bonding topology remain unchanged. A residual overlay is appropriate only when a named observable has a measured pull or an explicitly defined hypothetical model value. A phase diagram would be appropriate only if a deviation were large enough to reorder relevant levels or bonding topology. Present precision measurements strongly constrain such scenarios; they are included here as representation logic, not as expected physics.

## 11 Conclusion

An antimatter periodic table is not currently supported as a new chemical periodic law, and CPT symmetry gives no reason to expect one. It is useful as a CPT-isomorphic evidence-status and preparedness framework: it makes visible that antihydrogen is experimentally well anchored, antihelium-nucleus evidence is presently nuclear rather than neutral-atomic, and the wider anti-atom and antimatter-chemistry landscape remains largely open.

The scientifically robust formulation is:

We propose not an alternative periodic law for antimatter, but a CPT-isomorphic evidence-status and preparedness framework that represents orbital structure, experimental evidence, open chemistry, astrophysical search domains, and possible residuals as distinct layers.

The framework is intended as a conservative representation layer for current evidence and future searches, not as a claim of experimentally established antimatter chemistry beyond antihydrogen-related systems. Its value lies in separating structural expectation from experimental evidence, not in predicting a distinct antimatter chemistry.

## Acknowledgements

This work was partially supported by AI-assisted tools for structuring and drafting.

## Code Availability

This repository release is version v1.0.0 and is released without a DOI. A DOI-bearing archival release may be prepared separately in the future.

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